

Polina Baron

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Academic Appointment

Postdoctoral Fellow | University of Toronto (St. George), Department of Mathematics 2026–present
Supervisor: Kasra Rafi.

Education

University of Chicago, Department of Mathematics (Chicago, IL) Ph.D. 2020–2026
Advisor: Simion Filip. M.Sc. 2020–2022
Dissertation: “Semisimplicity, Rigidity, and Unique Ergodicity in Products and
Branched Covers of Translation Surfaces”.
Amick Fellowship.

Higher School of Economics, Department of Mathematics (Moscow, Russia) M.Sc. 2020–2022
Advisor: Aleksandra (Sasha) Skripchenko B.Sc. 2016–2020
Thesis: “Ergodic Properties of Linear Involutions”.
Academic scholarship for excellent students. Graduated Summa Cum Laude (“s otlichiem”).

Research Interests

My research is in dynamical systems and ergodic theory, with an emphasis on Teichmüller dynamics and translation surfaces. I study Hodge-theoretic rigidity of the Kontsevich–Zorich cocycle, unique ergodicity of branched covers, and rigidity phenomena for infinite-type translation surfaces and interval-exchange transformations. Additionally, I work on integrable systems, particularly the KdV hierarchy and its algebraic-geometric solutions. I am also interested in connections between translation surfaces and quantum chaos.

Publications and Preprints

P. Baron and E. Shuvaeva. **Unique ergodicity of branched covers of translation surfaces**. arXiv:2606.05492, 2026.
P. Baron. **The Neumann–Moser dynamical system and the Korteweg–de Vries hierarchy**. arXiv:2402.18079, 2024.
P. G. Baron. **The Mumford dynamical system and the Gelfand–Dikii recursion**. *Functional Analysis and Its Applications* 57 (2023), no. 4, 279–287.

Work in progress: Semisimplicity and rigidity of the Kontsevich–Zorich cocycle for products of strata.

Selected Talks

Institute for Advanced Study. Seminar talk. Princeton, NJ. May 6, 2026
Dynamics Student Days. Short talk. CUNY Graduate Center, New York, NY. May 4–6, 2026
University of Chicago. Seminar talk. Chicago, IL. February 16, 2026
Northwestern University. Seminar talk. Evanston, IL. January 27, 2026
Joint Mathematics Meetings. Washington, DC. January 4–7, 2026
Three short talks in the AWM Special Session on Research Developments by Women in Mathematical Physics, the AMS Special Session on Geometric, Dynamical and Probabilistic Aspects of Group Theory, and the AMS Contributed Paper Session on Dynamical Systems and Ergodic Theory.
Stony Brook University. Seminar talk. Stony Brook, NY. November 21, 2025
Semi-Annual Workshop on Dynamical Systems and Related Topics. Short talk. Pennsylvania State University. November 13–16, 2025
Semi-Annual Workshop on Dynamical Systems and Related Topics. Short talk. University of Maryland. April 4–7, 2024
Équations différentielles motiviques et au-delà. Online seminar talk, Institut Henri Poincaré (France). March 18, 2024

Additional Research Experience

Research Intern | Microsoft Research, Machine Learning and AI group (New York, NY) Summer 2024
Worked on an “astral space” approach to convex optimization at infinity. Studied convergence of sets and convex functions and the behavior of gradient and stochastic-gradient methods when minimizers escape to infinity. Mentors: Miroslav Dudík and Robert E. Schapire.

Research Intern | Institute for Mathematical and Statistical Innovation (Chicago, IL) Summer 2023
Served as the project's primary mathematical-methods consultant. Developed and applied statistical physics, persistent homology and topological data analysis methods to study criticality in deterministic and probabilistic simulations of functional brain connectivity for Alzheimer's disease research. Mentor: Alex Leow.

Junior Research Assistant | Higher School of Economics (Moscow, Russia) 2018–2021
Studied the ergodic properties of foliations, flows, and linear involutions on surfaces. Supported by joint RFBR–CNRS grant #18–51–15010.

Teaching Experience

Course Instructor | University of Toronto Winter 2027 (scheduled)
MAT136H1S: Calculus II
Scheduled to teach a lecture section in a coordinated, multi-section course.

Course Instructor | University of Toronto Fall 2026
MAT244H1F: Introduction to Ordinary Differential Equations
Independently design and deliver three hours of lectures per week for approximately 200 students. Prepare original lecture notes and course materials for every class; hold office hours; and collaborate with other instructors on assessment design and course administration.

Course Instructor | University of Chicago Fall 2024
MATH 112: Studies in Mathematics I (Introduction to Number Theory)
Designed and taught a 20-student introduction to number theory, including all lectures, course materials, assignments, and assessments.

Course Instructor | University of Chicago 2023–2024
MATH 151–153: Calculus I–III
Independently designed and taught all three courses in the sequence to approximately 30 students per quarter; responsible for lectures, course materials, assessments, office hours, and grading.

Lecturer | University of Chicago 2022–2023
MATH 131–133: Elementary Functions and Calculus I–III
Taught approximately 30 students per quarter in a coordinated, multi-section, three-quarter sequence. Prepared and delivered lectures following the shared syllabus; held office hours; and administered and graded common assessments.

Teaching Assistant | University of Chicago 2021–2022
Real Analysis I, Inquiry-Based Learning in Real Analysis II, ODEs, Brownian Motion.
Led problem-solving sessions, held office hours, and graded coursework.

Teaching Leadership and Academic Service

Phoenix STEM Collaborative Learning Coordinator | University of Chicago 2025–2026
Coordinated half of the tutorial sections for the 400-student MATH 131–132 calculus sequence. Observed tutorial instructors and provided pedagogical feedback, co-developed tutorial materials, and completed training in collaborative-learning pedagogy.

Dynamics Pre-Seminar Organizer | University of Chicago 2025–2026
Co-organized a graduate pre-seminar in dynamical systems, including selecting topics and coordinating speakers.

Organizer, Warm-Up Program for New Graduate Students | University of Chicago September 2024
Organized a week-long academic orientation program for incoming mathematics graduate students, including scheduling sessions and coordinating presenters.

Senior Mentor and Tutorial Coordinator | University of Chicago Mathematics REU Summer 2022
Mentored undergraduate researchers and supported their final presentations; coordinated tutorials, developed materials, and supported tutorial leaders.

Technical Skills, Languages, and Interests

Technical: $\text{\LaTeX}$, Python (NumPy, SciPy, Matplotlib, scikit-learn), SageMath, Wolfram Mathematica.

Additional expertise: Computational dynamics, convex optimization, statistical machine learning, topological data analysis, AI safety.

Languages: Russian (native), English (fluent), French (technical reading).

Interests: Creative writing; co-author of the first-place story in the 2021 *Novaya Fantastika* speculative-fiction contest (approximately 500 participants).